

Machine Learning Assignment 2

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REPORT

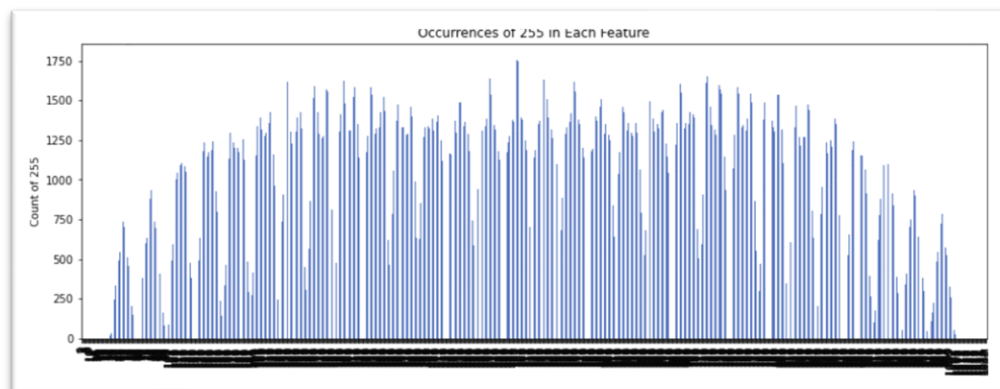
Algorithm used: Convolutional Neural Network

Design Decisions

Data Preprocessing

Initially the data consisted of 1041 features. Through observation it was found that the last column was strictly values 0,1, 2, 3. We observed the correlation between this feature and the output no correlation was found so it was removed for later use

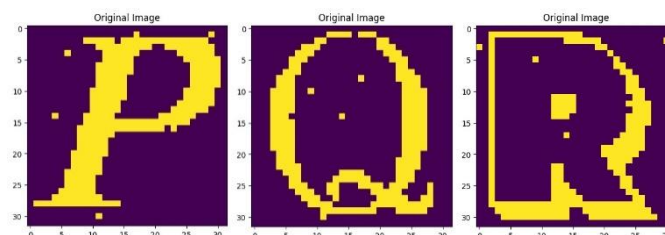
After plotting the data, it was found most values lay between 0 and 255 with 255 occurring in 17% of the data.



It was also found that a range of features had noise that skewed values over 255 some as large as 2000. 16 features had negative values throughout all the data, these negative features were noted and removed, and the outliers were scaled to a value between 0 and 255.

What remained was 1024 features which corresponds to a 32 by 32 image, and it was found that the last feature of the initial data referred to orientation hence we rotated the image and transformed it back into values for use in training the model.

Here are a few examples of the results



It is now apparent that all the data represents letters and each of the classes is a different letter.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	B	C	E	F	G	I	J	L	M	N	O	P	Q	R	S	T	U	V	W	Y

The orientations given in the last columns are as follows:

- 0: Normal Orientation (0 degree rotation)
- 1: 90 degree rotation anticlockwise
- 2: 180 degree rotation anticlockwise
- 3: 270 degree rotation anticlockwise

Model

This CNN model uses multiple convolutional (*the process of applying a mathematical operation that combines two functions to produce a third function to the input data*) layers to extract features from 32x32 grayscale images. Layers include:

Conv2D Layer: helps in detecting features such as edges, textures, etc., from the input images.

Batch Normalisation Layer: normalises the activations of the previous layer at each batch, which helps in speeding up training and provides some regularization to the model.

MaxPooling2D Layer: performs down-sampling (*reducing the dimensionality of the feature maps*), which helps in reducing the computational cost and controlling overfitting.

Dropout Layer: randomly sets a fraction of input unit to 0 at each update during training time, which helps in preventing overfitting.

The layers above are repeated 3 times with increasing filter sizes (32, 64, 128) to progressively learn more complex features from the input images.

Flatten Layer: flattens the 2D feature maps into a 1D vector that can be fed into fully connected (*Dense*) layers.

Dense Layer: applies a fully connected layer which combines all the features and contributes to the decision-making process of the network.

Final Dense Layer: produces the output predictions.

The model is trained with data augmentation and uses techniques like batch normalization and dropout to improve generalization and prevent overfitting. The model's performance is evaluated on both validation and additional test datasets, ensuring its robustness.

Hyperparameters

Some hyperparameters we used in the model include:

Number of filters: (32,64,128); Filter size: 3x3, Activation function: ReLU; Pooling size: 2x2; Dropout rate: 0.25; Number of neurons: 256; Activation function: ReLU; Optimizer: Adam; Loss function: Categorical Cross entropy; Number of epochs: 50.

Expected Accuracy: 80%